

ReLoop AI

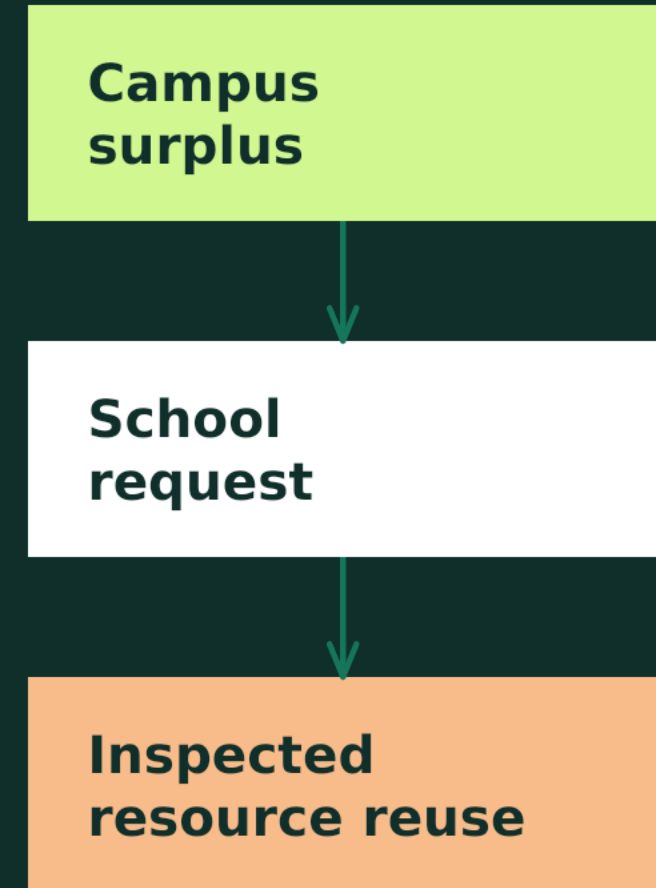
**Campus resources.
A second life in education.**

Campus-to-School Resource Reuse Network

RAG guidance + agent-assisted reuse planning

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A usable monitor can miss its next classroom

TARGET USERS

CAMPUS STORES & PROCUREMENT STAFF

Surplus equipment sits outside the purchasing workflow.

SCHOOL COORDINATOR

“We need two displays for our computer lab.”

How might AI connect reusable campus stock with school needs so institutions can avoid unnecessary purchases?

SDG 12 leads the project

12

Responsible Consumption and Production

Keep usable equipment in service.
Measure accepted reuse and mass.

04

Quality Education

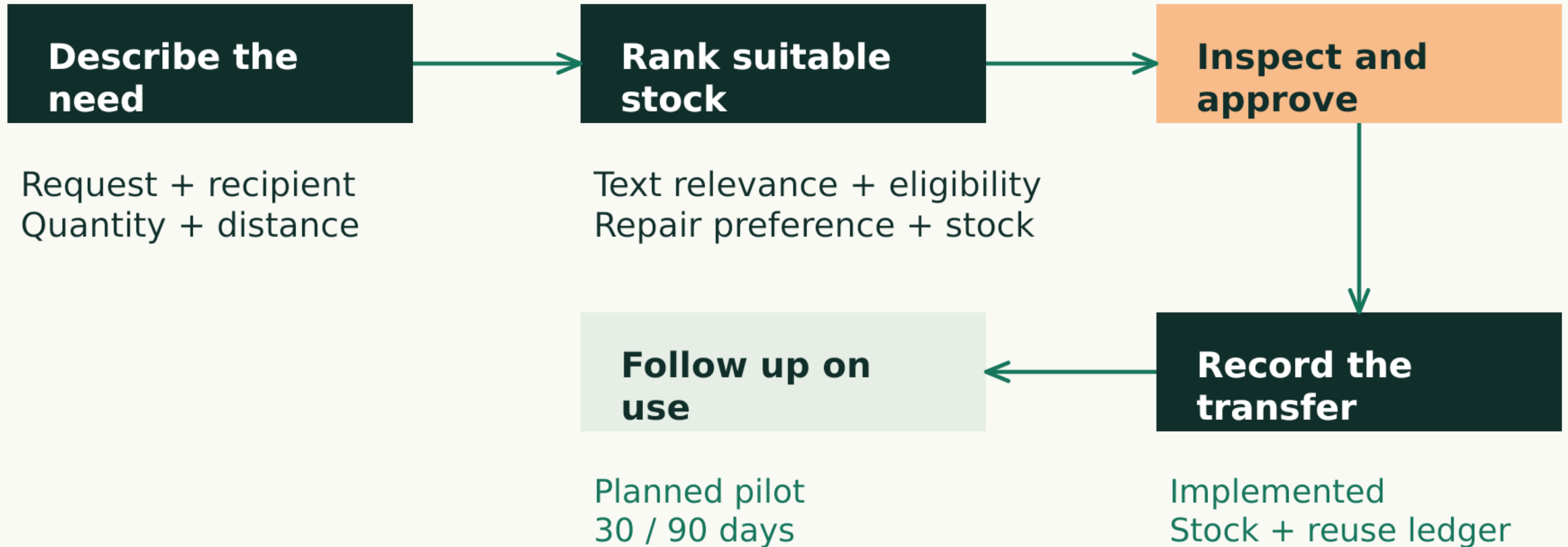
Equip learning spaces.
Check continued school
use.

17

Partnerships for the Goals

Enable school sharing.
Track active partners.

A reuse loop with people at the decision point



AI retrieves candidates. Rules check eligibility.

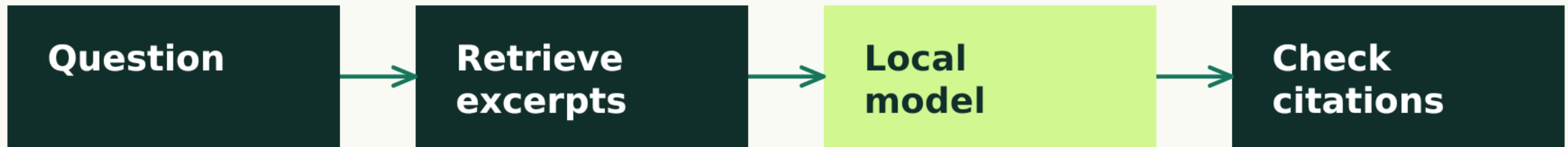
THE REQUEST

**“display for
computer
lab”**

Supported synonym:
display = monitor

- 01 Understand the wording**
Normalize supported terms before search.
- 02 Rank descriptions**
TF-IDF and cosine similarity score inventory.
- 03 Apply practical constraints**
Check recipient, distance, stock and repairs.
- 04 Explain the recommendation**
Show matched terms for human review.

RAG grounds answers in reuse guidance



Knowledge base

Project-authored rules, impact methods and responsible AI guidance. Heading chunks + TF-IDF retrieval.

Answer with evidence

Generated claims include source IDs. Missing evidence triggers abstention. A person checks what the excerpts support.

The agent chooses tools within fixed limits

MODEL PLANNER

**Select a tool.
Observe its
result.
Choose the next
step.**

Up to 6 turns / 8 calls

01 Search inventory

Use the user's fixed recipient and constraints.

02 Retrieve guidance

Bring inspection and reuse rules into the plan.

03 Estimate reuse

Calculate a draft from eligible, observed stock.

No transfer tool. The user must inspect and confirm.

Two modes, clearly labelled

OLLAMA MODE

Generative RAG Model-selected tools

Requires Ollama and a local model.
Questions and context stay
on-device.

NO-MODEL PREVIEW

Retrieved excerpts Scripted tool sequence

Runs immediately with Python.
Clearly labelled as a preview.

31 automated tests pass. Live-model quality and latency still need validation.

Running the full AI assistant

01 Install and start Ollama

02 `ollama pull qwen3:4b`

03 `python3 web_app.py --ai-mode ollama`

Open Reuse assistant. Ask about inspection, then prepare a reuse plan.

Setup: docs/AI_SETUP.md • No API key required

The prototype finds reusable resources

The screenshot displays the 'reloop AI' Resource exchange interface. The left sidebar contains navigation links: 'Resource exchange' (highlighted), 'Reuse impact', and 'Partner network'. Below these is a section titled 'A little more circular.' with statistics (04, 12, 17) and the text 'Education. Reuse. Partnerships.' and a user profile for 'Rishabh Bansal'.

The main content area is titled 'Workspace / Resource exchange' and features a 'LOCAL DEMO' button. The header reads 'REUSE BEFORE YOU REPLACE' and 'Find its next chapter.' with the subtitle 'Connect resources ready for reuse with people ready to use them.' A counter shows '13 available listings'.

A green banner states: 'One campus's surplus. Another classroom's possibility. Search approved demo inventory, review a match, and record a transfer. CAMPUS → SCHOOL'.

The search bar contains 'display for computer lab' and a 'Find matches' button. Below the search bar are filters: 'RECEIVING INSTITUTION' (set to 'Demo Community School'), 'QUANTITY' (set to '2'), and 'WITHIN DISTANCE' (set to '10 km'). A checkbox 'Include repairable items' is checked.

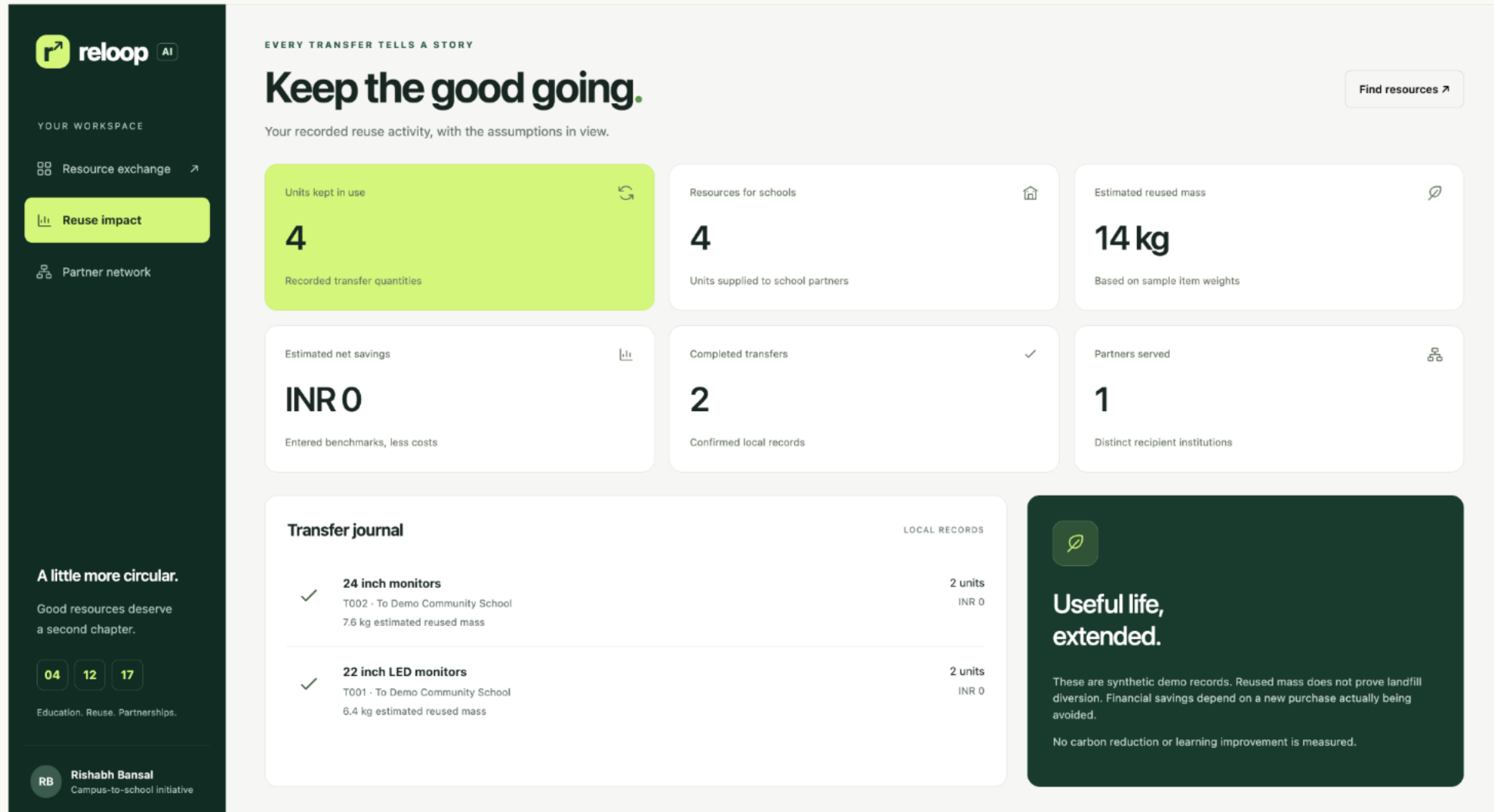
Below the filters is a 'Try a resource' section with tags: 'Monitors', 'Seating', 'Desks', 'Whiteboards', and 'Notebooks'.

The results section is titled '2 matching resources' and is 'RANKED BY TEXT RELEVANCE'. It shows two resource cards:

- 22 inch LED monitors** (ELECTRONICS TOP TEXT MATCH): 6 available, 3.4 km away, Working condition. Location: Computing Block A.
- 24 inch monitors** (ELECTRONICS MATCH 02): 1 available, 3.8 km away, Working condition. Location: Admin Block.

School request: 2 displays within 10 km. This view follows earlier demo transfers.

The ledger records each demo transfer



4 units reused. Estimated mass: 14 kg. Two transfers. Displayed savings: ₹0.

The network tracks partner eligibility

The screenshot displays the 'Partner network' section of the reloop AI workspace. The interface is divided into a dark sidebar on the left and a main content area on the right. The sidebar contains the reloop AI logo, navigation links for 'Resource exchange' and 'Reuse impact', and a highlighted 'Partner network' button. Below these is a quote: 'A little more circular. Good resources deserve a second chapter.' followed by three circular progress indicators with numbers 04, 12, and 17, and the text 'Education. Reuse. Partnerships.' At the bottom of the sidebar is a user profile for Rishabh Bansal, 'Campus-to-school initiative'.

The main content area has a header 'Workspace / Partner network' and a 'LOCAL DEMO' button. Below this is the heading 'BETTER, TOGETHER' and the main title 'Different places. Shared possibilities.' with a subtitle 'The institutions in this fictional campus-to-school network.'.

Three institution cards are displayed:

- Demo Campus**: CAMPUS - CAMPUS01, Approved demo partner (green status).
- Demo Community School**: SCHOOL - SCHOOL01, Approved demo partner (green status).
- Demo School awaiting review**: SCHOOL - SCHOOL02, Awaiting review (orange status).

Below the cards is a green section titled 'One project. Three connected goals.' with three columns:

- 04 Quality education**: Help schools access suitable equipment for teaching. Intended contributions. No real institutional agreements or measured outcomes yet.
- 12 Responsible consumption**: Keep useful materials in service before buying new.
- 17 Partnerships for the goals**: Coordinate accountable sharing between institutions.

Two approved demo partners and one awaiting review. All institutions are fictional.

What makes this approach distinctive?

01

Reuse before purchase

Begin with a real equipment need, then search existing stock.

02

School-aware matching

Apply recipient eligibility and school-sharing permissions.

03

A traceable transfer

Connect the match to approval, stock updates and reuse estimates.

Future concept: a resource passport with repair history and reuse follow-up.

Responsible AI is part of the workflow

Fairness

Test varied vocabulary. Review missed needs.

Transparency

Show sources, tool activity and model limits.

Privacy

Use fictional data. Collect no student details.

Ethics

Agent proposes. A person inspects and confirms.

A working baseline with a clear test plan

13

synthetic listings

9

supported item types

31

automated tests passed

NEXT: VALIDATE WITH USERS

Interview campus and school staff. Confirm the visibility gap.
Label 50 requests and measure useful matches.
Refine the prototype with mentor feedback.

From one local demo to an institutional network

NOW

Local prototype

Single user
Synthetic partners
JSON stock ledger

PILOT

Campus + school

Verified stock
Partner agreements
Human transfer review

SCALE

Shared network

Transactional database
Roles and reservations
Procurement
integration

Impact means equipment that stays useful

10

accepted transfers

Proposed pilot target

30 / 90 days

Continued-use checks

Environment

Extend item life and avoid some new purchases.

Education

Supply usable equipment to learning spaces.

Institutions

Build a repeatable sharing process.

ReLoop AI

**A reuse check before
the next purchase.**

github.com/rishabh230707-afk/ReLoop-AI

Browser demo: `python3 web_app.py`

School example: `python3 reloop.py school-demo`

Sources: internship brief, project code and UN SDGs 4, 12, 17